

Macro for CS: Group Project Instructions

The group project receives a total of 40 points, which corresponds to 40 % of the final grade. The points are split between the presentation and the submitted materials, i.e., presentation slides, supporting material, and computer codes.

Group formation:

You should form groups of three students and self-allocate to one of the eight groups created on the course's Canvas/StudyNet page, which is named "Project_Group" number X.

Topic choice:

Each student should choose one country, except the United States of America, from the list of 185 featured in the Penn World Table version 11. The country choice should be indicated in a discussion that will be open on the course's Canvas/StudyNet page starting March 20 at noon. Countries are allocated on a "first-come, first-served" basis, and no country can be chosen by more than one student.

Submission:

The slides for the presentation must be submitted electronically in PDF format, on the course's page by selecting the assignment "Group Project" and uploading the file, along with the computer codes and any other supporting materials.

The deadline for the submission is 10 p.m. on May 21, 2026. You are allowed multiple submissions within the submission period. The last submitted version will be graded. No submissions are allowed after the deadline.

Presentation:

Each group should deliver an oral presentation in class. The total presentation time is 15 minutes, equivalent to 5 minutes per country. Each group member should do a part of the presentation.

Content:

For each country, you should collect data, derive stylized facts, implement and simulate the models studied in class, report and compare the results. In particular, you should:

Part 1 - Economic Data:

a) Collect data on the variables discussed in Lecture 2 (e.g., nominal GDP, real GDP, GDP deflator, CPI, population, unemployment rate, etc.) for the chosen country. It is preferable to use quarterly series, but if not available, then annual data is acceptable. The longer the time series, the better it is for your analysis. Some possible data sources:

- Penn World Table version 11.00, only annual data, available at <https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/FABVLR>
- World Development Indicators dataset, available at <http://databank.worldbank.org/data/reports.aspx?source=world-development-indicators>
- Eurostat dataset, mainly for European Union countries, available at <https://ec.europa.eu/eurostat/web/main/data/database>
- Federal Reserve Economic Data, available at <https://fred.stlouisfed.org>

- OECD Data Explorer, mainly for OECD countries, available at <https://data-explorer.oecd.org>
- b) Construct the series for real output per capita. Plot the series and discuss its evolution over time. Construct and plot the growth of real output per capita. What is the average growth rate over the whole sample? What can be inferred about the country's business cycles?
 - c) Plot and discuss the evolution of the unemployment rate over the whole sample. What is the average unemployment rate? What are the largest and the smallest values, and when were they reached? Are there any particularities of the labor market in this country?
 - d) Construct and plot the inflation rate. Discuss the evolution of the inflation rate in this economy. Is monetary policy inflation targeting? If so, since when and how successful has the monetary authority been at ensuring price stability?
 - e) Using the stylized facts discussed in Lecture 3 as a benchmark, create similar stylized facts for your chosen country and discuss them. For example, some questions to be answered are:
 - i) Is the country among the rich or the poor countries?
 - ii) Has it experienced high or low growth over the past decades? etc.

Part 2 - Model Simulation:

- a) Take the general Solow model studied in Lecture 4 to perform a model simulation for the chosen economy. You may choose any programming language you prefer (e.g., MATLAB, Python, R) for writing the code.
- b) Use the data in the Penn World Table version 11.00, available at <https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/FABVLR>, for your country, to calibrate the model parameters - the capital share of income α , the population growth rate n , the depreciation rate of capital δ , the output per capita growth rate g , and the investment share of GDP s . Assume that A is initially equal to one, i.e. $A_0 = 1$.
- c) Assume that the country is in a steady state, i.e., that the per capita variables grow at a constant rate (balanced growth) from $t = 0$. Compute steady-state values for the variables.
- d) Consider that in period 20 there is an exogenous shock that permanently changes one of the parameters of your choice, e.g., s , n , etc.. Motivate the parameter choice for the policy exercise and the magnitude of the change.
- e) By what percentage do the steady state values change after the permanent shock?
- f) Create time series for $\tilde{k}_t$, $\tilde{y}_t$, $\tilde{c}_t$, A_t , k_t , y_t , c_t , $\ln(y_t)$, $\ln(c_t)$, and $g_t^y = \ln(y_t) - \ln(y_{t-1})$. Simulate 100 periods.
- g) Create a set of diagrams showing the evolution over time of the above-mentioned variables and discuss your results.

Part 3 - Countries Comparison:

Perform a comparison of the three countries. Discuss similarities and differences and possible factors, in particular government policies, that may explain them.